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'''
LP-III Assignment 3 (Group A - DAA)
Write a program to solve a fractional Knapsack problem using a greedy method.
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class Item:
    def __init__(self, profit, weight):
        self.profit = profit
        self.weight = weight
def fractionalKnapsack(w, arr):
    arr.sort(key=lambda x: x.profit/x.weight, reverse=True)
    finalValue = 0.0
    for item in arr:
        if w >= item.weight:
            finalValue += item.profit
            w -= item.weight
        else:
            finalValue += item.profit * (w/item.weight)
            break
    return finalValue
if __name__ == "__main__":
    n = int(input("Enter number of items-\n"))
    arr = []
    for i in range(n):
        profit = int(input("Enter profit of item " + str(i + 1) + "-\n"))
        weight = int(input("Enter weight of item " + str(i + 1) + "-\n"))
        arr.append(Item(profit, weight))
    w = int(input("Enter capacity of knapsack-\n"))
    print("Maximum value in knapsack: ", fractionalKnapsack(w, arr))
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OUTPUT:-

Enter number of items-

3

Enter profit of item 1-

60

Enter weight of item 1-

10

Enter profit of item 2-

100

Enter weight of item 2-

20

Enter profit of item 3-

120

Enter weight of item 3-

30

Enter capacity of knapsack-

50

Maximum value in knapsack: 240.0

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